Self-Exercise: FHS, Disks, Filesystems, and Archives

For the successful completion of the tasks, it is enough to use just one machine. Perhaps, a new one.

The only special requirement is to have an additional (second) disk of 2GB.

Let's assume that the following list of tasks will be performed on a new stand-alone machine:

1. Create archive of **/etc** with **xz** utility and **measure** the **time** needed. Name the archive **etc.tar.xz**
2. Create archive of **/etc** with **bzip** utility and **measure** the **time** needed. Name the archive **etc.tar.bzip**
3. Create archive of **/etc** with **gzip** utility and **measure** the **time** needed. Name the archive **etc.tar.gzip**
4. Compare the **size** of the **three** archive files created earlier. **List** their **names** and **sizes** (using the **ls** command) ordered from smallest to the largest and store them in a file **compression-test.txt**
5. Partition the disk using **GPT**:
    a. **First** partition of type **Linux** and **700MB** in size;
    b. **Second** partition if type **Linux Swap** and **200MB** in size;
    c. **Third** partition of type **Linux** and **300MB** in size;
    d. **Fourth** partition of type **LVM** and **100MB** in size;
    e. **Fifth** and **sixth** partitions of type **LVM** and **300MB** in size;
6. **Create** folder **/addon** and subfolders (check bellow) to mount the new filesystems
7. Create **xfs** filesystem on the **first** partition, mount it at **/addon/xfs**, and include it in the **/etc/fstab**
8. Initialize the **swap** partition, **turn it on**, and include it in the **/etc/fstab** file
9. Create **ext4** filesystem on the **third** partition, mount it at **/addon/ext4**, and include it in the **/etc/fstab**
10. Create **PVs** on the **fourth**, **fifth**, and **sixth** partitions
11. Create new **VG** named **vg_addon** and include all **three PVs** created earlier
12. Create new **LV** named **lv_addon** with size **400MB**
13. Create **ext4** filesystem on the new **LV**, mount it at **/addon/lvm**, and include it in the **/etc/fstab**
14. Extend the **LV** to include all the space available on the **VG**
15. Extend the **LV's filesystem** to include all the available space
16. **Reboot** the system and **check** if everything with the new filesystems is okay (**list them**).

** Please note that all resulting files should be in the home folder of the user used for the homework solving process*

Self-Check

After you are done with the tasks, you can check how well you did them

In a terminal session execute one of the following commands (depending on which executable you have)

- Option 1 (**wget** installed, usually for **Debian**-based distributions)
  wget -q https://courses.zahariev.pro/m6.sh -O - | sudo bash
- Option 2 (**curl** installed, most of the distributions)
  curl -s https://courses.zahariev.pro/m6.sh | sudo bash

Repeat the procedure until you get as many **PASS** marks as possible. This may include adjustments on what you did or even start from the beginning

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